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Palaeomagnetism of the Permian volcanic rocks of Moissey (French Jura): implications for the palaeofield and tectonic evolution

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Summary. Rocks from the Massif de la Serre in the French Jura (latitude: 47.3°N longitude: 5.6°E) belonging to an ignimbritic assemblage dominated by vitrophyric rhyolites, and whose age of formation is probably Permian (Autunian to Saxonian) have been studied by applying thermal and alternating field demagnetization. The characteristic magnetization has a mean direction derived from 89 samples of $D = 170°$, $I = -16°$, $k = 26.2°$, $\alpha_{95} = 3°$ and a corresponding north palaeopole at 41°N, 172°E, $A_{95} = 5°$. The pole, which is very close to the Permian European poles, can thus be considered as a new contribution. Some samples are found to carry a unique normal polarity magnetization, others carry both normal and reverse polarities. It therefore seems that, similar to Permian series in the USSR, these west European rocks have registered a normal event in the Kiaman interval. From a structural point of view, we may conclude that during the Alpine tectonic phases the Massif de la Serre has not been subjected to substantial rotation.

Key words: palaeomagnetism, reversals, Permian, volcanics, Jura

1 Introduction

Situated at the western margin of the Jura, the Massif de la Serre corresponds to a horst whose granites and gneisses form the only outcrop of the Franco-comtois basement (Fig. 1). This horst is associated with sediments of Palaeozoic age some of which have yielded a Permian flora. A particular siliceous rock, called 'eurite' (see below), outcrops in the north-western part of the massif, between the gneiss and the Palaeozoic sediments. Several origins have been suggested for this rock: metamorphic, sedimentary, volcanic and volcano-sedimentary (see accounts by Bonte 1943; Devaux 1960 and Zogo 1981). The most recent interpretation (Thiebault & Jacquin 1961) again takes up the volcanic origin hypothesis,

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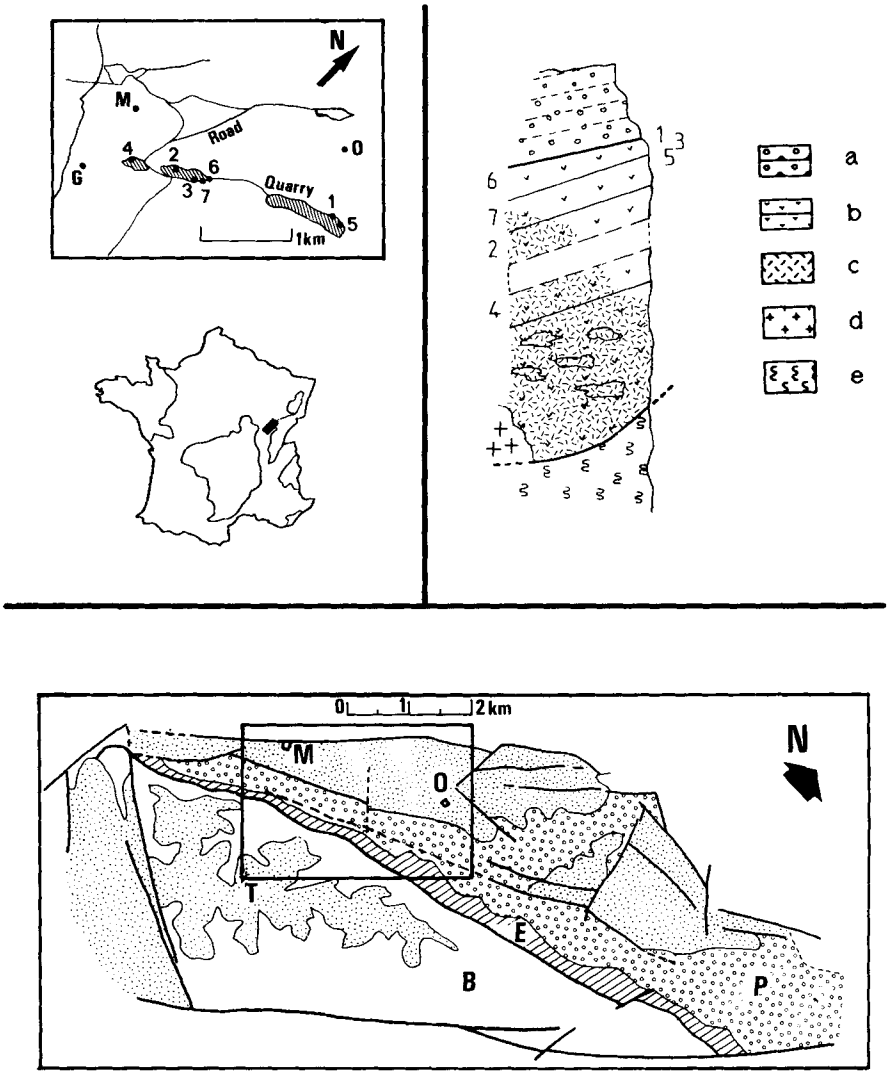


Figure 1. Geological sketch map of the Massif de la Serre (from Zogo 1981) and sampling sites (situation plan). M: Moissey, O: Offlanges, G: Moissey gorges, B: basement, E: eurite, P: Permian sediments, T: Trias, 1–7: sampling sites (two groups in rectangle outline around quarries). Top right: synthetic log of the Palaeozoic rocks of the Massif de la Serre (partly from discussions with J. Dalsass). (a) Palaeozoic sediments; (b) green eurite; (c) red eurite; (d) granophyre; (e) gneiss; 1–7: sampling sites.

considering the eurite as a devitrified volcanic tuff and noting its similarity with the Permian ignimbrites ('welded tuffs') of the Nideck (Vosges). We shall see later that there is indeed a rhyolitic and ignimbritic assemblage here, and it was hoped that a study of the magnetic remanence carried by the rocks would allow new information to be obtained on the palaeomagnetic field in Europe during the Permian. Furthermore, conclusions of a general interest, not only from the point of view of palaeomagnetism (definition of a Permian pole) but also on the evolution of the Jura within the Alpine context, may be drawn from the study.

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2 Geological setting: nature and age of the eurite

2.1 BASEMENT

The Serre horst includes a section raised through the sedimentary cover (Fig. 1); to the south-east it is in contact with the Jurassic, and to the north-west with the volcanic assemblage and the Permo–Carboniferous. To the south-west it is unconformably overlain by Triassic sandstones.

The relation between the granito-gneissic massif and its substratum is unknown. In particular, present data do not show whether an intracrustal shear plane (low seismic velocity level, which appears to exist in the Alps) exists in the Jura, above which the Serre block might have been subjected to décollement. It will be shown below that the palaeomagnetic data furnish information on this question.

2.2 PERMIAN SEDIMENTS

Two main groups have been distinguished in the Palaeozoic sediments: at the base, above a breccia containing elements of volcanic rocks, are ochre-coloured sandstones with highly coloured breccia or clayey levels and layers of brown silt and micaceous sands. This detritic assemblage outcrops in the Moissey gorges; there, the micaceous levels have yielded a flora associated by Corsin (in Devaux 1960) with the upper Autunian or lower Saxonian; at the top, lying unconformably on its substratum, a red detritic unit (sands and conglomerates) attributed in the literature to the Saxonian.

2.3 VOLCANIC ROCKS (EURITE)

The term 'eurite' was created in the last century to denote any unit formed by rhyolites and microgranites. Fallen into disuse in the recent literature, it has however been used until now to denote the ambiguous rocks of the Serre.

The eurite forms a continuous band which obliquely traverses the Massif de la Serre from south-west to north-east; it is separated from the basement by a fault (Fig. 1) which appears to be subvertical at the surface, but which has been shown by boreholes to be inclined and to flatten with increasing depth (mean dip of about 40° to the NNE; concavity towards the top).

This silica-based, stratified rock is at least 15 m in thickness and shows coarse grained layers towards the base, while towards the top the texture is compact and hard, mostly fine grained, of vitreous appearance, where it is industrially exploited as ballast or chippings.

The eurite contains undeformed cubes of pyrite which were probably formed during the consolidation of the rock. In the lower part of the formation the colour of the rock is reddish and the pyrite crystals are mostly replaced by haematite. In the upper part of the formation the colour is generally grey-green, the pyrite being very often fresh.

In the Moissey gorges, the eurite outcrops in the form of a layer considered by Bonte (1943) as a conglomerate. The fault mentioned above forms its contact with the gneiss situated further east. The contact with the Permian fossil-bearing formations is not visible here; the two formations are perhaps separated by a fault at this point.

In the quarrying area around Moissey-Offlanges, the green eurite ends in a hardened surface which obliquely slices the stratification and evens out in some small north-easterly directed faults. The detrital layers which rise above the eurite over this surface are therefore slightly discordant to their substratum (a few degrees); these red-brown formations have some green argillaceous lenticular levels. The facies are different from those of the gorges

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series; they could as easily be of Saxonian as of Autunian age. Lienhardt (1962), however, has shown a section to the east of Offlanges where the sedimentary terrains, whose facies are equally different to those of the gorges and are attributable to the Saxonian, could also lie unconformably over the eurite.

In conclusion, the attribution of the eurite to the Permian, Autunian to Saxonian, appears justified within the local context as well as on a regional scale (Alps, Vosges).

Radiometric measurements have been made by J. M. Cantagrel using K-Ar techniques: the age furnished by this study is 209 ± 4 Myr, which corresponds to the Triassic period. However, the Trias is represented here by light-coloured sandstones, very similar to those well known in the Vosges; these sandstones overlie the Permian facies sediments, themselves underlain by the green eurite of the Offlanges quarry. This absolute age is therefore a minimal age only, which is not unexpected for a vitric rock for which argon loss is usual during devitrification.

2.4 VULCANOLOGY

The workfront of the Offlanges quarry in 1983 June showed an upper level of micropheric green eurite (originally glassy, but devitrified) at the top, and with coarser grains lower down. Within this coarse facies occur submicroscopic elements consisting of 'corroded' quartz, alkaline feldspars, and fragments of gneiss and quartzites. These elements also exist in the finer grained facies, but more rarely. The thickness of this upper level diminishes towards the SE (at most 4–5 m), the upper surface of the eurite being oblique to the stratification.

In the Offlanges quarry, this level lies above a coarser level which locally has conglomeratic facies (elements of gneissic rocks). This lower level is very silicified, above all in the south-eastern part of the quarry.

These two units, which dip at approximately 20° to the NW, have been found throughout the zone exploited by the quarries at Moissey, which would imply an extension of several kilometres to the SW. The lower level alone was noted by Bonte (1943) in the old Moissey gorges quarry; he found there 5 per cent of fine grained silicified eurite amongst the elements of this formation.

The Moissey quarries (Fig. 1) do not lend themselves so well as those of Offlanges to vulcanological observations. We may however note the presence of joints which probably correspond to limits of old prisms at the NE extremity of the quarry, which at present contains the processing plant (site 7).

The Serre eurite thus corresponds to a group of rocks belonging to an ignimbritic and rhyolitic assemblage predominated by vitrophyric rocks similar to those described in the southern Alps (Haut Adige, in Mittemperger 1960). The particular granulometry of the vitreous levels would seem to indicate a rapid cooling (probably in lacustrine conditions), at least of the external parts of the flows. The presence of crushed granophyres (of the Lugano type) is not however excluded in the south-eastern part of the quarry, where they must until now have been grouped with the gneiss of the basement. Furthermore, certain facies of the lower level are doubtless comparable to those of rheoignimbrites (as described by Rittman 1958) or perhaps of sillars (as described by Pichler 1963, p. 416).

It is possible that the inclined fault separating the eurite from the gneiss was one of the fissures along which the magma flowed. The Permian sediments and the eurite have evidently become detached from the granito-gneissic basement at this contact level during the Tertiary tectonic phases. The silification observed along the feature probably corresponds to impregnations of silica-bearing hot water; and the red colour observed throughout the lower

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levels to oxidations, *per ascensum*, of fumerolian origin (Fig. 1). This point is important since it impinges on the mineralogy of the remanent magnetization carriers. In the case of haematitized red rocks, there is always the possibility that the alteration was very late. Here, we may consider that the red levels, being close to the contact with the basement, have undergone alteration very soon after their formation.

An important question for the palaeomagnetist is whether the volcanic rocks studied were cooled on an horizontal or an inclined surface. As is indicated by the regional structural context, it is logical to assume that the eurite has, like the Permian sediments, been tilted during the formation of the Serre horst. However, the nature of the eurite-gneiss contact, and the eurite inclusion facies, are compatible with cooling on an inclined surface. The results will be presented below with and without correction for a dip of 18° bearing 15°W.

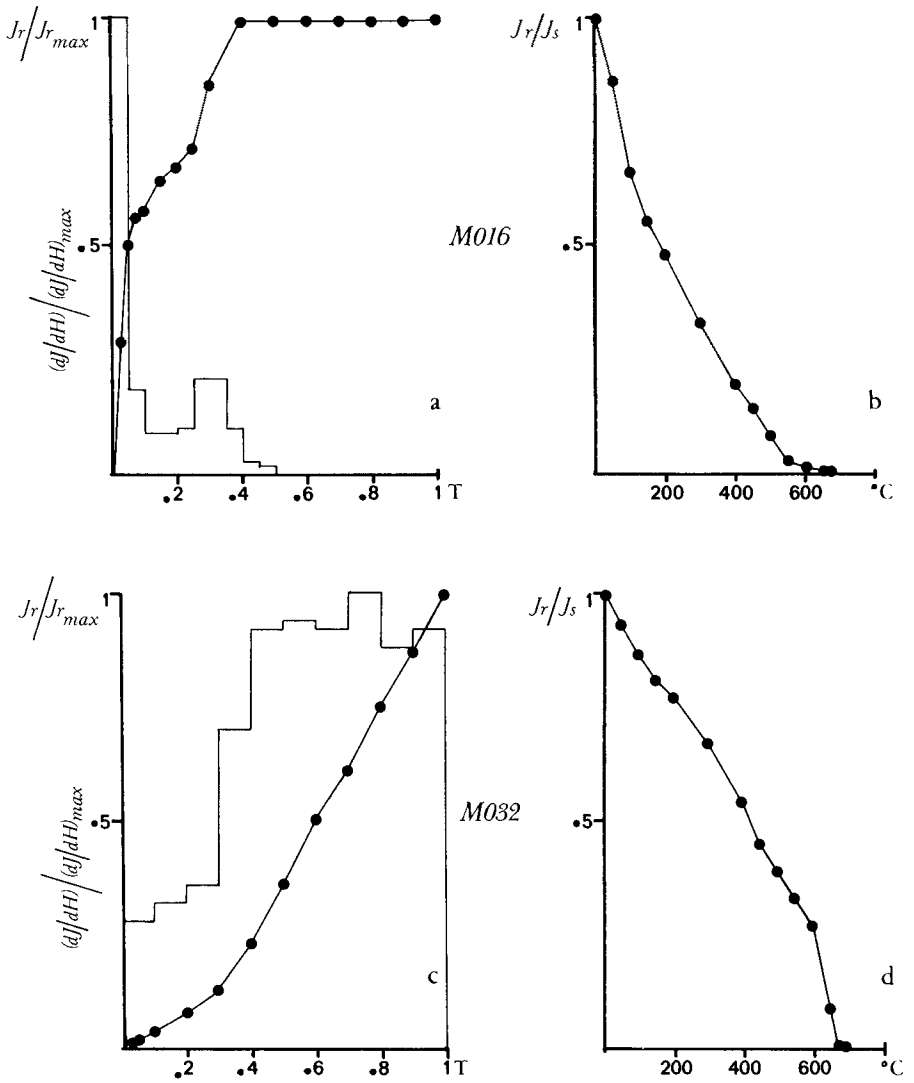


Figure 2. Typical IRM acquisition curves (a, c) and thermal demagnetization curves (b, d) of the maximum IRM, for the green unit (above) and red unit (below).

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3 Sampling and measurement procedures

Taking into account the restricted nature of the access to the formation, more than 10 samples were collected from each of the seven sites either by coring or more often (owing to the hardness of the rock) by hand sampling using plaster of Paris caps (Thellier 1938) for orientation. Sampling was carried out on both the upper unit (Fig. 1) whose grey-green tint would appear to be the original colour, and the lower unit, where the red colour posed a problem as noted above (Section 2). The sampling strategy was aimed at obtaining not only a good horizontal but also a good vertical distribution.

In the laboratory at least three specimens were cut from each sample, so as to enable systematic magnetic (alternating field) and thermal cleaning, sometimes the two techniques being employed together on the same specimen (Irving & McGlynn 1975; Daly 1981), and also low field anisotropy measurements.

The remanent magnetization carried by the specimens was measured using principally a

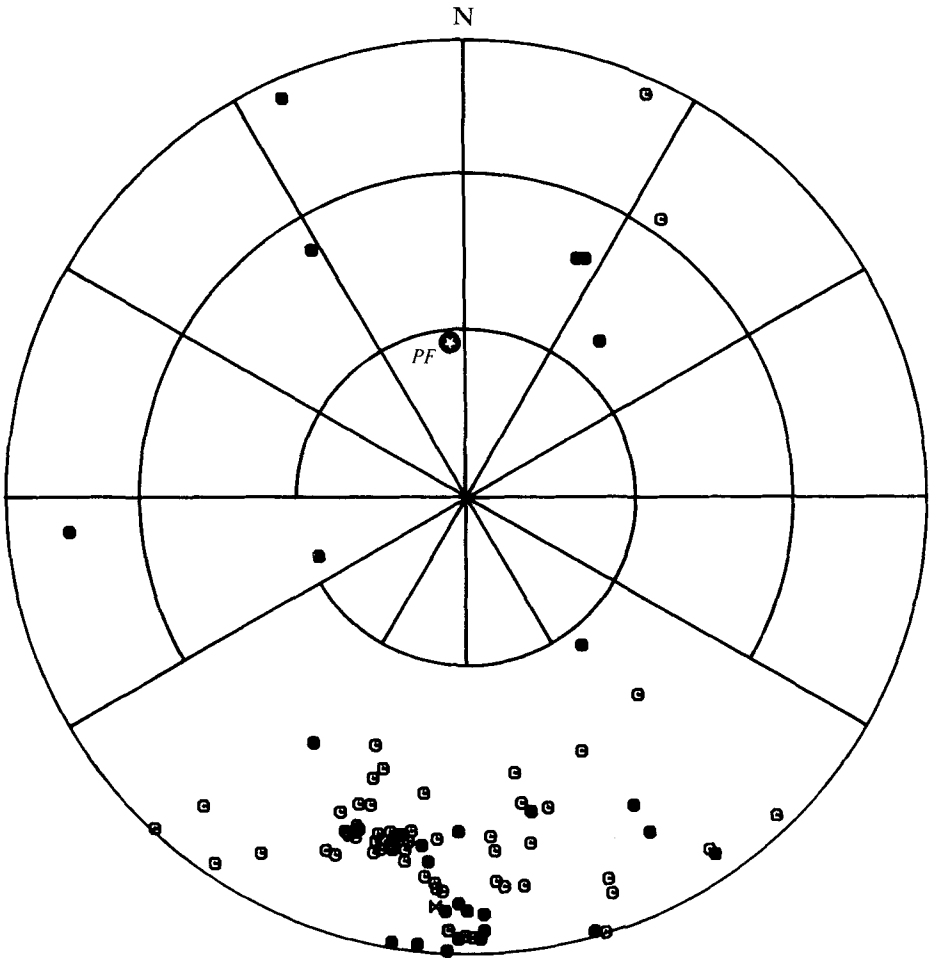


Figure 3. NRM directions of the 89 specimens (selected after treatment) plotted on the whole sphere of an equal-area projection. Filled or open symbols represent vectors pointing downwards or upwards respectively. PF is the direction of the Earth's field at present.

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JR-4 spinner magnetometer (Geophysika, Brno) and additionally a 3-axis RS-01 cryogenic magnetometer (LETI/CEA) in the Laboratoire de Stratigraphie of the University of Paris VI. Newly developed apparatus was used for magnetic and thermal cleaning. The furnace used is a single-chamber type with a five-layer mu-metal cylindrical shield and forced air cooling. The AF apparatus is characterized by automatic cancelling of the anhysteretic effect in fields up to 280 mT (Le Goff 1985). Treatment steps adopted varied according to the specimen response but were usually 5 mT for magnetic cleaning and 100°C up to 500°C and then finer treatment (50°, 25°) for thermal demagnetization.

The results were analysed mostly using classical methods such as linear sections in orthogonal vector plots, or by defining stable grouping of magnetization vectors during magnetic and thermal cleaning on equal-area projection nets. In addition, the multivariate technique of principal component analysis (Kirschvink 1980) was used in cases where complex configurations were confused by noise; and in particular for the definition of the best fit plane containing subtracted vectors and corresponding maximal and minimal variance.

Anisotropy measurements were realized by B. Henry using a CRAD Digico spinner at the Laboratoire du Centre des Faibles Radioactivités at Gif sur Yvette. Coercivity spectrum analysis by acquisition of isothermal remanent magnetization (IRM) using an electromagnet producing 1.5 T, and subsequent thermal demagnetization was carried out as a means of characterizing the magnetic carriers (Lowrie & Heller 1982).

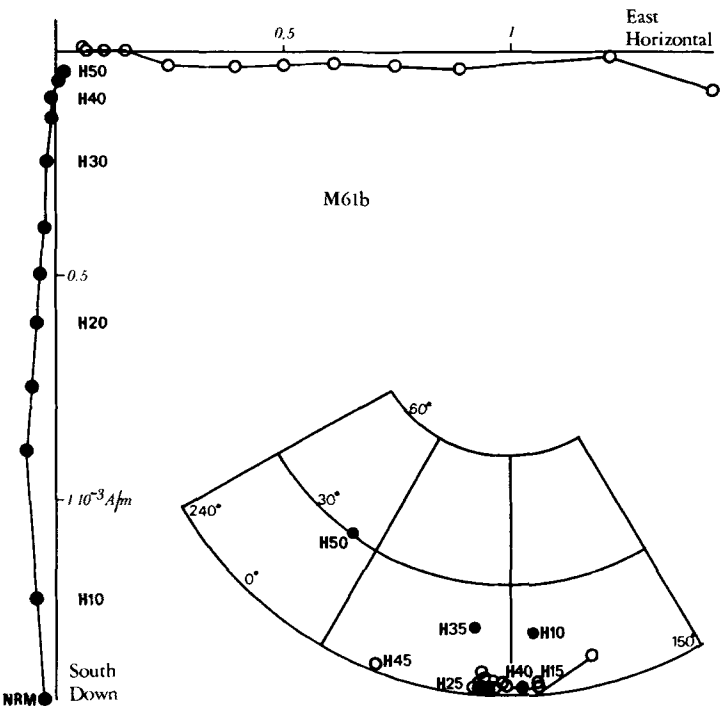


Figure 4. Case type A. On an equal-area plot a single reverse polarity component is defined by close grouping of remanent (symbols joined by a line) and subtracted vectors. The latter are identified by their treatment code: H = alternating field (values in mT). For this case the orthogonal vector plot shown is that using a vertical variable plane passing through successive remanent vectors giving the true inclination for the vertical plot (open circles). Filled circles are used for the horizontal plot.

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4 Magnetic behaviour

The magnetic viscosity was evaluated on a representative set of samples by two measurements; the first made after a prolonged stay in zero field, and the second after a period of two weeks in the terrestrial magnetic field (TMF), the specimens being positioned such that the z-component was parallel with the field vector in the laboratory (Thellier & Thellier 1959; Prévot 1981). The viscosity index, expressed as the ratio of the change in the z-component to the moment of the natural remanent magnetization (NRM), is variable; low in cases where haematite is the principal carrier (of the order of 0.05) but in the upper unit much higher values were recorded, so that a zero field environment was systematically used during experiments and for storage.

A study of the magnetic susceptibility anisotropy (Graham 1954; Daly 1970; Hrouda 1982) was made on a series of specimens drawn from all sampling points. The results showed firstly the weakness of the anisotropy, secondly an important dispersion of the principal axes and finally that there was no evidence for either foliation or lineation, as indicated by the parameter $f(\approx 0.5)$: we define f by: $f = (ka - kb)/(ka - kc)$ where the principal susceptibilities $ka > kb > kc$ (Henry & Daly 1983). The weakness of the anisotropy, as well as the scattering of the axes, shows that the Serre eurite has probably not been subjected to significant deformations. The weak anisotropy therefore reflects, as in the case of alpine

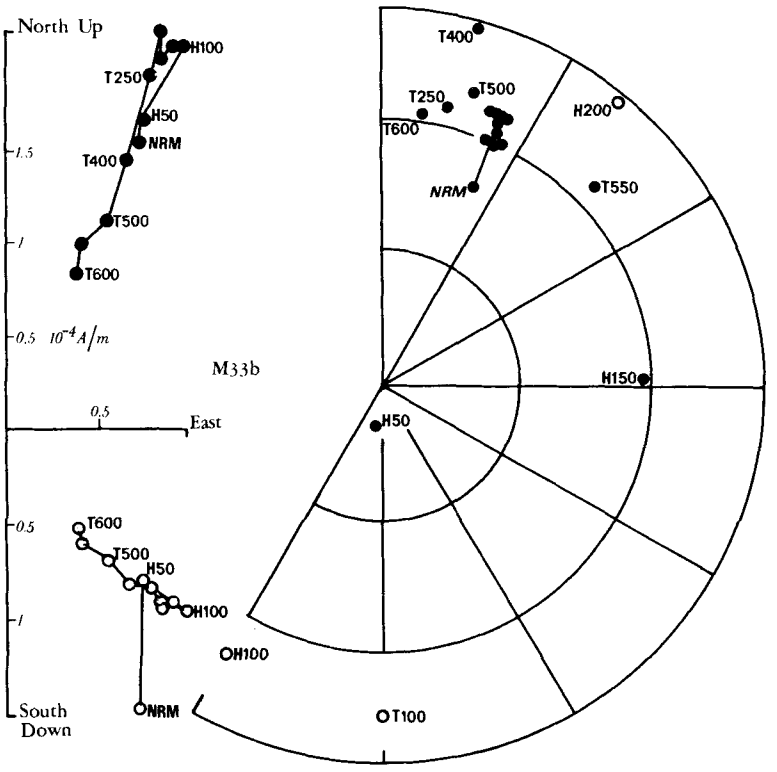


Figure 5. Case type B. Equal-area plot shows a dominant normal polarity component defined by closely grouped remanent and subtracted vectors to the NNE; the presence of another component is indicated by some SSW subtracted vectors. T = temperature (in °C). For this and the following cases a conventional vector plot is used.

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lava flows (Henry 1980), the flow structure of the rock, and we find that after correction for geological dip the principal axes may be fitted to a vertical plane of symmetry dipping at 87° towards the SSW, which would appear to contain the direction of the volcanic rock's flow, which would therefore be WNW–ESE.

Coercivity spectrum analysis by acquisition of isothermal remanent magnetization (IRM) and subsequent thermal demagnetization (Horner & Heller 1983) was used for identifying the ferromagnetic content of the grey-green and red coloured samples. These experiments were made on a representative set of samples in progressively increasing DC fields up to 1 T and results are illustrated in Fig. 2. The IRM acquisition curve of grey-green samples shows two populations of low coercivity minerals, one of which reaches saturation around 0.1 T being probably magnetite. It is difficult to say if the other population, reaching saturation in higher fields, is another phase of magnetite or whether it is due to another mineral such as pyrrhotite (Fig. 2a). Progressive thermal demagnetization of the saturation IRM (Fig. 2b) gives unblocking temperatures around 550°C supporting the presence of magnetite in the grey-green samples. The IRM of red samples is characterized by a broad range of coercivities (Fig. 2c), saturation being clearly unattained at 1 T. The IRM thermal demagnetization curve (Fig. 2d) presents some irregularities around 180°C suggesting a minor proportion of goethite, and around 400°–500°C implying the presence of titanomagnetites, but the major carrier is without doubt haematite whose presence was evident from the colour.

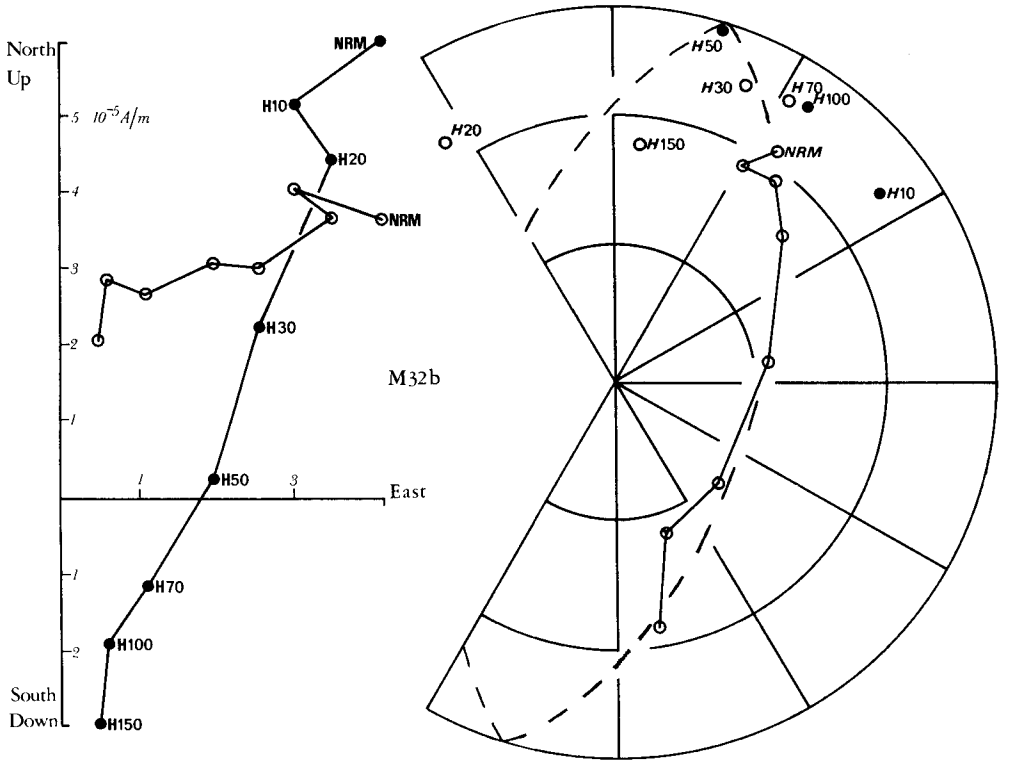


Figure 6. Case type C. On equal-area plot, gradual removal of normal polarity component is reflected by evolution of the remanent vector from the NNE through a sector of a plane towards the south. The orthogonal plot shows incomplete separation of the reversed component.

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5 Magnetic analysis

The natural remanent magnetization intensities are low, ranging from 0.5 to 45×10^{-4} A m^{-1} with a mean value of 14×10^{-4} A m^{-1}. The NRM vectors are for the most part directed towards the SSW, with relatively low positive and negative inclinations (Fig. 3), the mean direction being $D = 175°$W, $I = -11°$. This direction, obtained from measurements after storage in zero field is thus significantly removed from the viscous component. The scatter is, however, relatively wide and although the distribution is difficult to visualize because of the subhorizontal positions of the NRM vectors, the trend is found to be oblate in a north–south orientation. This may be better seen when the group is rotated through 40°–60° so bringing nearly all of the points into the same hemisphere. The north–south trend represents a first indication of the possible existence of multicomponent magnetizations.

Magnetic analysis of the specimens allows four main characteristic cases to be distinguished:

(a) The specimen carries a single or very dominant reverse magnetization shown by the resultant vector measured after each magnetic or thermal cleaning step. There is little or no significant variation in direction, and the subtracted vectors are closely grouped around the resultant vector position, indicating a stable end point (Fig. 4).

(b) The magnetization of the specimen is dominated by a normal component. The resultant vector obtained at each measurement is situated to the NE, but the specimen may also show signs of the presence of a second component, often characterized by (usually

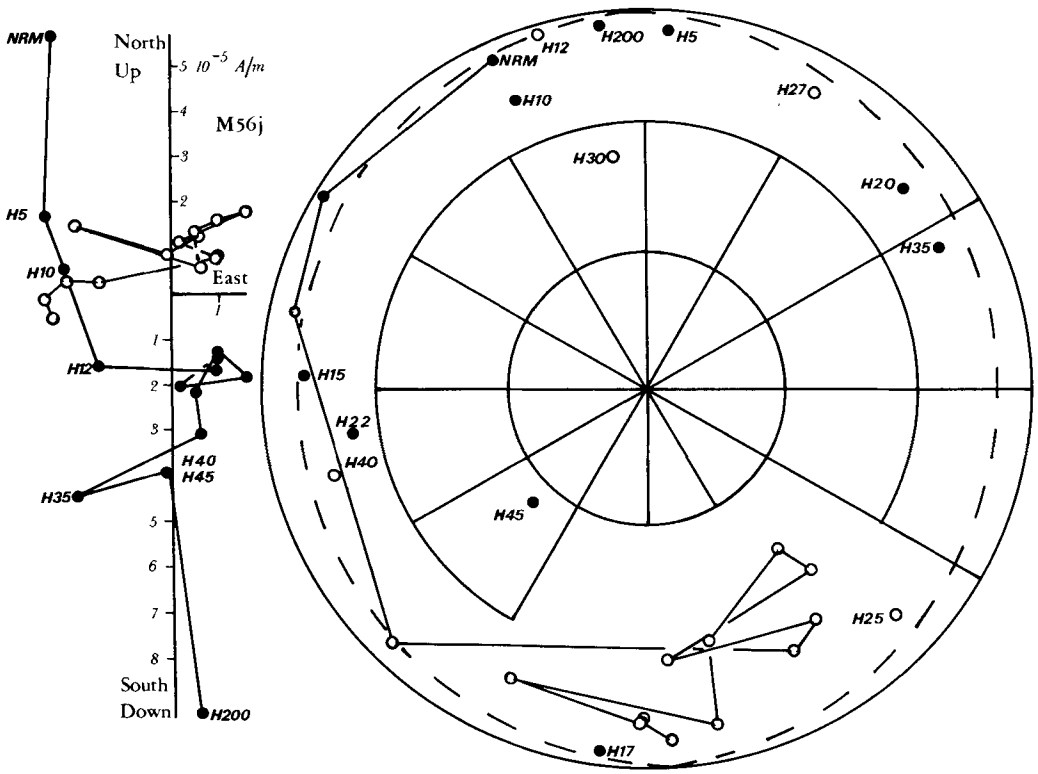


Figure 7. Case type D. On an equal-area plot, irregular evolution in a whole plane of remanent and subtracted vectors appears to be the result of one of the two components having registered fluctuations of the field. Irregular variations are also reflected in the orthogonal vector plot.

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reverse) subtracted vectors directed towards the SSW. The two components have very similar spectra for blocking field (H_b) and blocking temperature (T_b), which may produce some wandering of the resultant vector direction (Fig. 5).

(c) Two components also exist, but with different H_b or T_b spectra (Fig. 6). In the NRM the normal component is dominant, and directed towards the NNE. As magnetic cleaning continues, however, the normal component is gradually removed, as shown by the subtracted vectors clustered near the NRM direction, indicating a lower H_b than the reverse component, which finally becomes dominant. The evolution of the resultant vector during magnetic cleaning describes a sector of the plane in which the normal and reverse components lie, as shown in Fig. 6.

(d) In certain very weakly magnetized specimens ($J \approx 6 \times 10^{-5}$ A m^{-1}), the resultant vectors and subtracted vectors describe the whole plane containing the two magnetic components. For the example shown in Fig. 7, principal component analysis (Kirschvink 1980) has been used to calculate this plane from the subtracted vector directions. As in the previous case, the magnetic component with a normal direction has a lower blocking field, and is defined by the mean of the first three subtracted vectors, taken as non-unit vectors. Similarly, the reverse magnetic component is obtained from the resultant vectors at the end

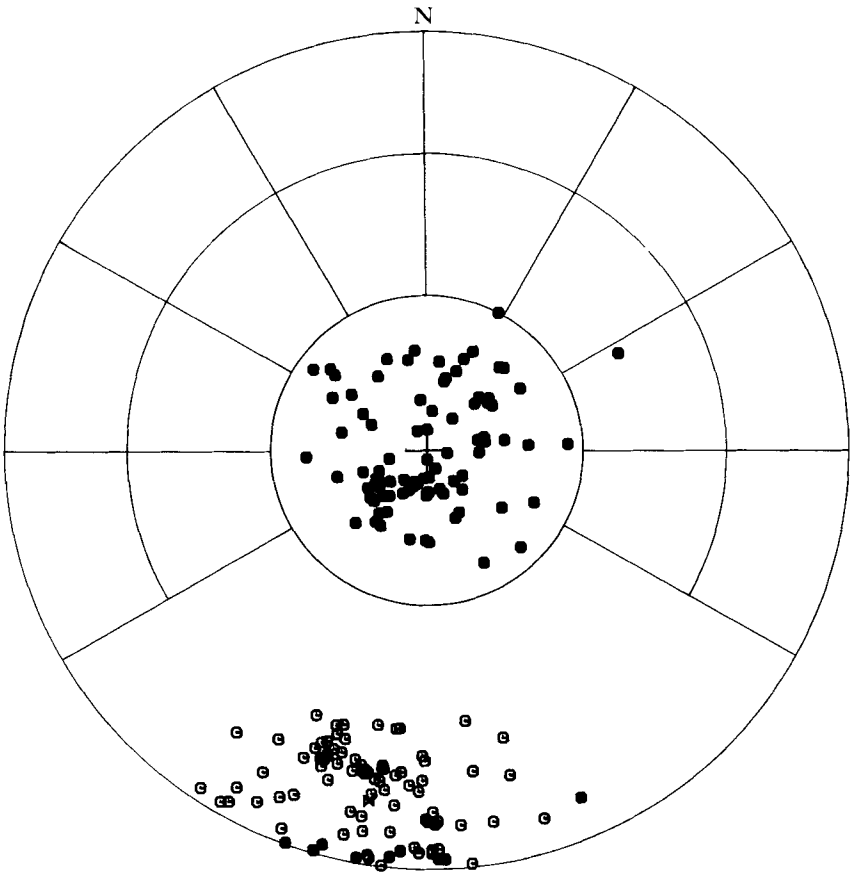


Figure 8. Reverse polarity directions of all specimen magnetizations obtained after magnetic analysis (normal vectors being inverted and plotted as reverse). At centre: the same distribution after rotation through 106° bringing the mean direction to the centre, so that the grouping may be better visualized.

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of the demagnetization path. The plane containing the two components shows a maximum variance about a mean direction fairly close to the calculated direction of the reverse component, while the pole of the plane is near the vertical, reflecting the subhorizontal nature of the two components. We may question the source of such significant dispersion of the vector points which are clearly within the whole plane, rather than a sector. While the magnetization is very low, other samples of similar intensity give clear single or two-component analyses, thus for the most part ruling out noise. A possible explanation is that we are seeing an effect produced by short-term variations in the direction of the TRM during and shortly after the formation of the rock.

6 Palaeomagnetic and geological conclusions

The demagnetization treatment described above resulted in a considerable improvement of the directional grouping (Fig. 8) particularly in cases where magnetite was found to be the principal carrier, thanks to the elimination of minor VRM components and in general a clear isolation of the characteristic remanent magnetization direction. The mean vector direction however did not change greatly, defined by $D = 170°\text{W}$, $I = -16°$ when giving equal weight

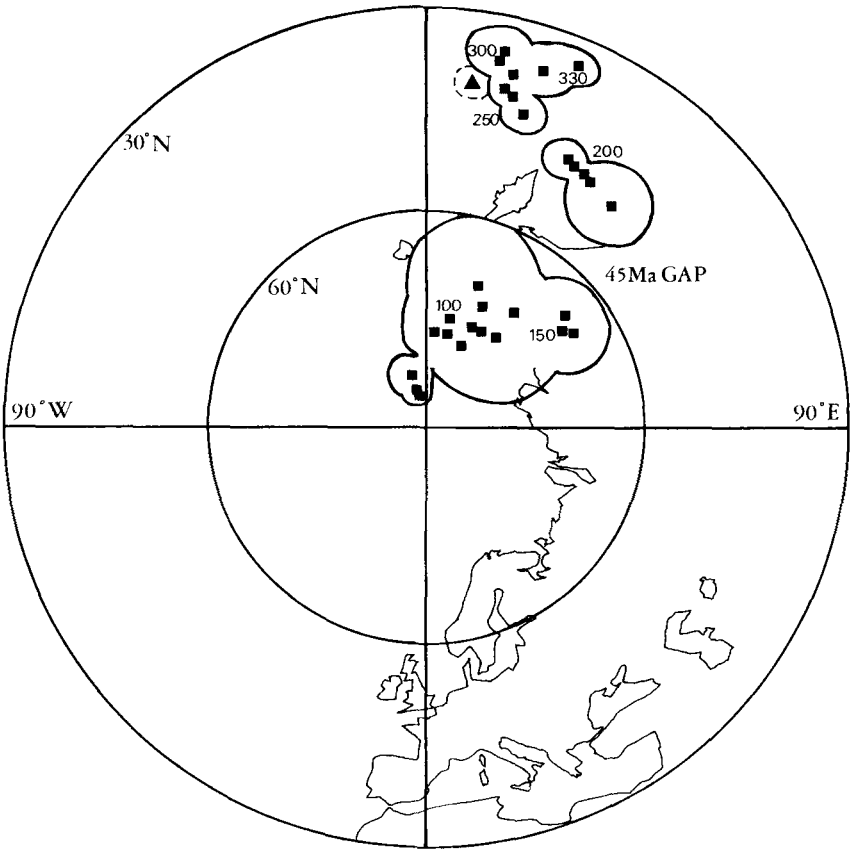


Figure 9. The apparent mean polar wander path for northern Eurasia (data from Irving & Irving 1982 using a 30 Myr sliding window). Squares mark the positions at 10 Myr intervals. Solid line represents external limit defined by 0.63 confidence circles associated with each pole. The triangle marks the Moissey pole position with its confidence circle. Points on the APW path with less than four source poles have been ignored. Northern Eurasia is shown in its present position. Equal-area projection.

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to individual specimens, with $k = 26.2$ and $\alpha_{95} = 2.9°$ ($N = 89$). Mean values from site to site were not found to differ significantly and the mean direction giving equal weight to each site (calculated from site means) $D = 169.2°$W, $I = -15°$ with $k = 80.9$ and $\alpha_{95} = 5.8°$ ($N = 7$), remains close to the direction obtained from individual specimens. For the purposes of these calculations, normal vectors obtained from certain specimens were inverted and then treated as reversed. These data allow the calculation of a palaeomagnetic pole with respect to the present position of the sampling site defined by $\lambda = 49.3°$N and $\Phi = 169.1°$E before dip correction and $\lambda = 41.3°$N, $\Phi = 172.4°$E after correction for a dip of 18° bearing 15°W; $K = 123.3$, and $A_{95} = 4.8°$. The position of this pole (after correction for dip) is shown in Fig. 9, against the apparent polar wander APW path for northern Eurasia as calculated by Irving & Irving (1982) using a 30 Myr window. As can be seen, this pole is very close to the Permian part of the north European APW path. The rather easterly position relative to the neighbouring mean pole positions (northerly relative to the Permian pole) may be due to a number of factors, such as block movement of the Serre, and the uncertainty over the true dip correction (if any) to apply. A further possibility is that, since the assumed primary magnetization is a TRM, acquisition occurred during a relatively short period thus recording a field direction whose pole was not coincident with the geographical pole, as a result of secular variation. We may also note that other discrete poles between 270 and 260 Myr record even more easterly positions. There is no doubt, however, that the new pole is situated within a critical part of the path at the beginning of the rapidly changing Permo–Trias section.

The analysis of the characteristic remanence carried by the specimens also allowed the isolation of a normally directed vector in a number of cases (Table 1). This vector is most readily observable in samples from the red-coloured lower unit (haematite carrier) where the resultant vector is strong and directed towards the NNE subhorizontally. In these samples this probably represents the primary magnetization while in others the normal polarity being defined by subtracted vectors during thermal demagnetization is likely to be due to partial

Table 1. Summary of results after magnetic analysis. N is the number of samples for each site, K the precision parameter. D_m, I_m, λ, Φ, α_{95}, A_{95}, A_{63} are respectively in degrees the declination and inclination of the magnetization, latitude and longitude of palaeomagnetic poles, the radius of the circle of confidence, and errors of the palaeopoles at 0.95 and 0.63 probability giving unit weight to each of the seven site poles. Sites coordinates (at Pesmes): latitude 47.3°N, longitude 5.6°E. Geological stratigraphic correction: dip of 18° bearing 15°W.

Sites	N	Dm(°)	Im(°)	α_{95}(°)	λ(°N)	Φ(°E)	K	A_{95}(°)	A_{63}(°)
M1	13	-170.8	-8.7	8.4	46.3	172.3			
M2	10	-160.3	-23.2	3.6	51.1	153.9			
M3	15	-178.4	-17.9	7.9	51.8	183.0			
M4	22	-167.3	-22.0	2.3	52.6	164.8			
M5	14	-170.6	-13.3	7.9	48.6	171.4			
M6	9	-174.7	-5.8	6.7	45.4	178.1			
M7	6	-162.0	-13.6	14.9	46.8	159.0			
all samples	89	-169.8	-16.1	2.9	50.0	169.8			
all sites									
before dip correction	7	-169.2	-15.0	5.9	49.3	169.1	123.3	4.8	2.4
after dip correction	7	-170.1	1.3	5.9	41.3	172.4	123.3	4.8	2.4
normal component									
before dip correction	19	14.0	10.7	9.8	46.4	165.2			
after dip correction	19	13.6	-5.1	9.8	38.7	168.1			

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remagnetization. We would therefore appear to have isolated a normal polarity magnetization acquired by some samples.

The presence of this normal polarity in rocks from the firmly established reverse Kiaman interval which extends from about 300 to 220 Ma is an interesting topic. A narrow zone of normal polarity has been reported at the base of the Autunian series of the USA (McElhinny 1969). This zone is also found in the USSR series where short normal polarity episodes also appear to have been identified near the top of the Permian. It would be particularly interesting if we were able to show the existence of a comparable zone for western Europe.

However, it is necessary to bear in mind the fact that this effect may be due to total or partial self-reversal. A number of authors have suggested ways by which a rock may acquire magnetization in a direction opposed to that of the contemporary field, ranging from self-reversal in a magnetic mineral such as magnetite (Néel 1951) to chemical changes where chemical differences would be observed in reversed and non-reversed rocks. These mechanisms have been summarized for example in van Zijl, Graham & Hales (1962), although very few cases of spontaneous reversal have been well established. While it is not definitely possible to prove or disprove self-reversal without further work on the ignimbrites from Moissey, the favoured interpretation would be that of a normal polarity field vector recorded soon after the formation of the rocks during the crystallization of haematite at the end of the Permian, as in the USSR formations.

In geological terms, the directional results set out above show good coherence from site to site, thus supporting the hypothesis proposed at the beginning of the study, that the rock was formed at high temperatures, being of volcanic origin. The flow direction suggested by the anisotropy study provides additional evidence in support of this interpretation.

From the structural point of view, we may conclude that if there was a décollement of the Massif de la Serre on its substratum during the Alpine tectonic phases, this phenomenon did not provoke any substantial rotation of the block relative to the direction of the Permian magnetization vector. The position found for the pole corresponds closely to that expected from other poles of similar date, and these results are coherent with an hypothesis of non-décollement of the Jurassic upper crustal level during the Alpine phases.

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